Riley Hernandez

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Technical Skills

Languages: Rust (tokio, serde, thiserror), Python (Flask, NumPy, Pydantic), TypeScript (Node.js, Next.js, React), F#, Gleam
Hardware: Onshape, 3D Printing (FDM, SLA), Circuit Prototyping, DFM/DFA
Other: Linux (Development, Server, Embedded), Cloud Infrastructure (GCP)

Professional Experience

Caldo Restaurant Technologies — Oakland, California
Founding Robotics Engineer — 01/2024 - Present

Embedded Controls & Sensing

- Designed and implemented real-time control systems in Rust for robotic food dispensers, using a proportional control architecture driven by multi-load-cell sensor fusion to achieve precise portion control under dynamic, high-vibration conditions.
- Created a calibration algorithm based on matrix test readings and LU decomposition, automatically determining sensor weights and offsets for accurate multi-load-cell performance across differing geometries and payloads.
- Developed signal-processing pipelines (low-pass filtering, moving-window smoothing) to mitigate noise and drift from electrical and mechanical sources.
- Managed process concurrency through multithreading, async runtimes, and actor-model channels for reliable I/O handling and deterministic control loop timing on ARM-based Linux SBCs.
- Implemented self-diagnostics and health checks for calibration consistency, sensor drift, and communication latency.

Cloud Architecture & OTA Systems

- Engineered bidirectional communication between devices and cloud via authenticated HTTP APIs on GCP.
- Built full OTA configuration-update infrastructure, enabling remote calibration, firmware settings modification, and device-specific tuning from the cloud.
- Developed TypeScript/React/Tauri-based tooling for commissioning, calibration visualization, and live telemetry graphing — streamlining setup and maintenance by field technicians.
- Designed structured data schemas and APIs for telemetry.
- Integrated cloud dashboards for data monitoring and analytics to guide service, QA, and restocking decisions.

System Integration, Mechanical Design, and Production

- Led internal mechanical design reviews, set redesign timelines, and coordinated with manufacturing partners for production readiness.
- Served as CAD manager and BOM owner in Onshape — maintaining scalable designs and organized version control.
- Personally designed critical mechanical components for food-contact modules, sensor mounts, and control panels emphasizing cleanability, modularity, and manufacturability (DFM/DFA).
- Trained mechanical team on Onshape-specific and general design process, version control, and vendor documentation.
- Oversaw transition from prototype to scalable production, managing vendor sourcing and production line setup in Costa Rica.
- Supported deployments in national and local QSR chains and corporate hospitality campuses, validating reliability and calibration stability in live environments.

Mezli — San Mateo, CA
Mechanical Engineering Intern — 06/2021 - 01/2023

- Designed and prototyped electromechanical systems for bowl position control using Onshape and Arduino
- Designed new and improved existing systems for increased functionality, meal and device aesthetics, food regulation standards, ease of use, and cleanability
- Conducted DOE for dispensing device's material and design to optimize aesthetic plating without food waste
- Coordinated with local and international suppliers and manufacturers for fabrication cost minimization
- Rapid-prototyped a novel automated denesting system for bowl dispensing utilizing compliance and a singular actuator

Tesla — Fremont, CA
Manufacturing Engineering Intern — 1/2022 - 05/2022

- Designed and prototyped pneumatically driven actuator for position control of car body skids using SolidWorks
- Designed and implemented new compressed air preparation system for pneumatic actuation and under body sealing material delivery to robotic dispensing arms for application
- Designed mechanical aspects of a gantry-mounted lidar system to take in and analyze point cloud data for car body skid quality inspection
- Coordinated with contractors for PLC and electrical cabinet design, installation, and incorporation into existing infrastructure

Education

University of California, Berkeley Berkeley, California
BS, Mechanical Engineering + BS, Business Administration 2020 - 2024

- Organizations: Hardware Lead @ Neurotech@Berkeley, Founding VP Projects @ Engineering Solutions at Berkeley
- Relevant Coursework: Mechatronics Design, Electronics for IOT, Computational Biomechanics, Entrepreneurial Leadership, Powers and Politics in Organizations, Managing Technological Innovation and Strategic Growth
- Other Coursework: Music Cognition & Perception, Music of Brazil, Race & Order in the New Republic (1800s American Literature), The Social, Political, & Ethical Environment of Business

Misc.

- Licenses & Certifications
  - ServSafe Food Handler
  - Teaching English as a Foreign Language (TEFL)
  - USA Archery Coach (Level One)
- Volunteering
  - ESL Teacher – Las Casas English Second Language Program
  - Archery Coach and Device Technician – Break the Barriers Disabled Veteran Program
- Hobbies
  - Musician
  - Rock Climbing
  - Cooking
  - Reading
  - Film